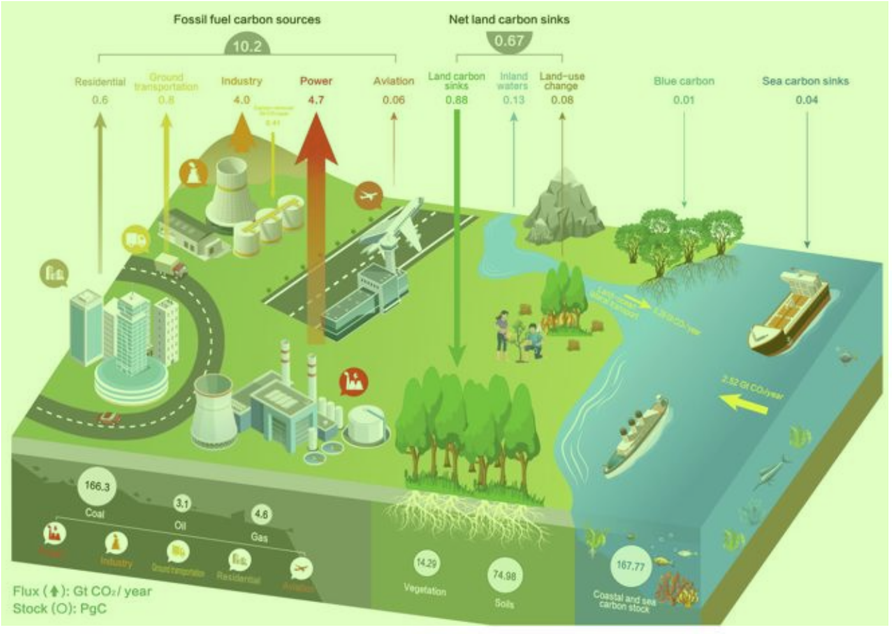
Near-real-time monitoring of global ocean carbon sink [1]

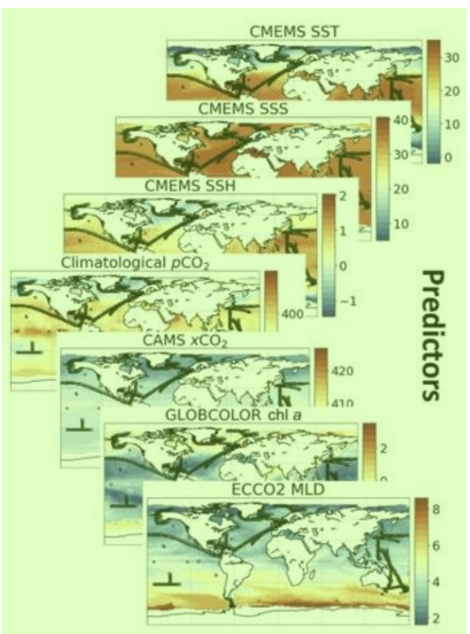
Piyu Ke, Xiaofan Gui, Wei Cao, Dezhi Wang, Ce Hou, Lixing Wang, Xuanren Song, Yun Li, Biqing Zhu, Jiang Bian, Stephen Sitch, Philippe Ciais, Pierre Friedlingstein, Zhu Liu [2]

Contact: xiaofangui@microsoft.com [3]

Introduction [4]


 [5]

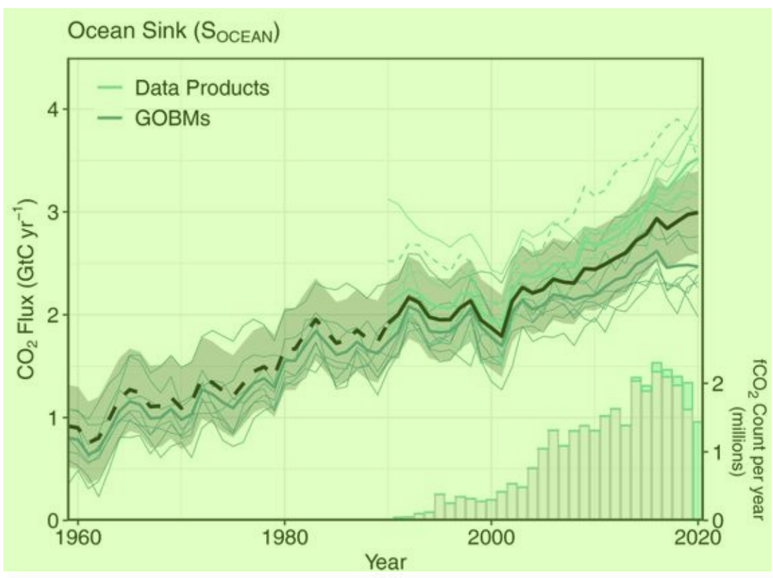
We present Carbon Monitor Ocean (CMO-NRT), a novel dataset providing near-real-time monthly gridded estimates of global surface ocean fugacity of $CO_2(fco_2)$ and ocean-atmosphere CO_2 flux from January 2022 to July 2023. Leveraging Convolutional Neural Networks and semi-supervised learning, our highly accurate models enhance timely climate change mitigation efforts. [6]


 [7]

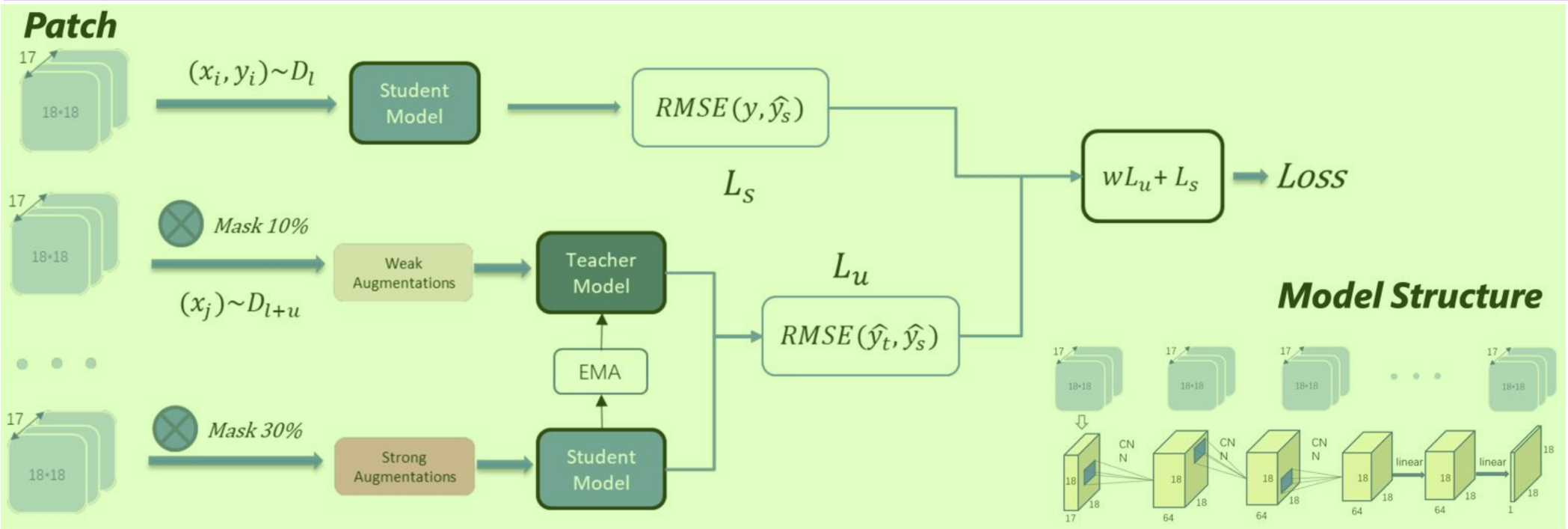
Environmental factors [8]
Predictors

Global ocean carbon sink
Target [10]

- **Environmental factors influencing and reflecting oceanic carbon sink variations**
- **Reconstructing global oceanic carbon sink using ML methods with global environmental data** [9]

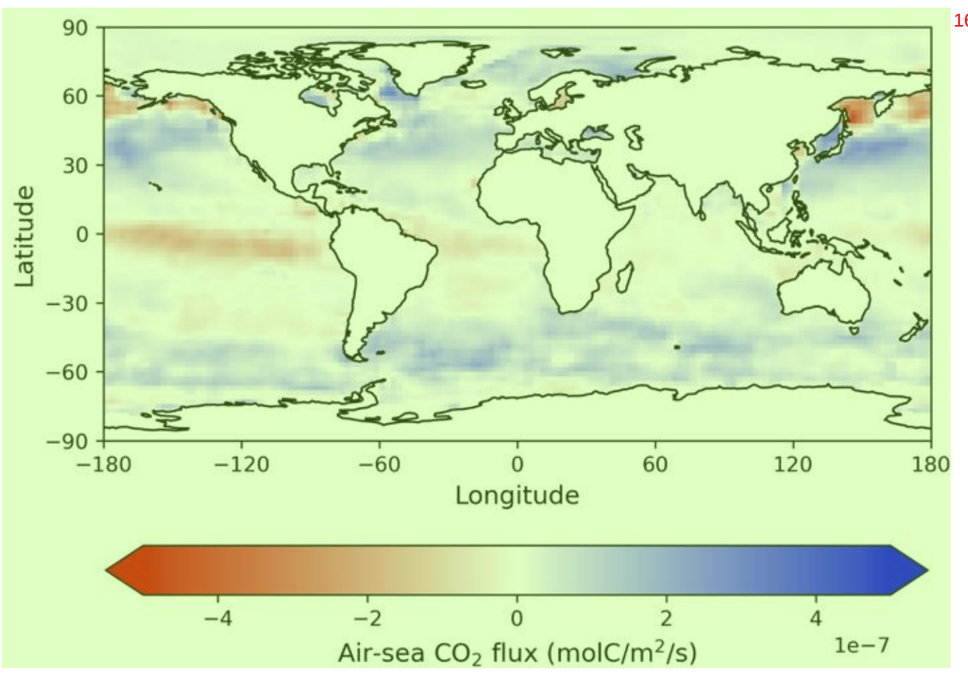

 [11]

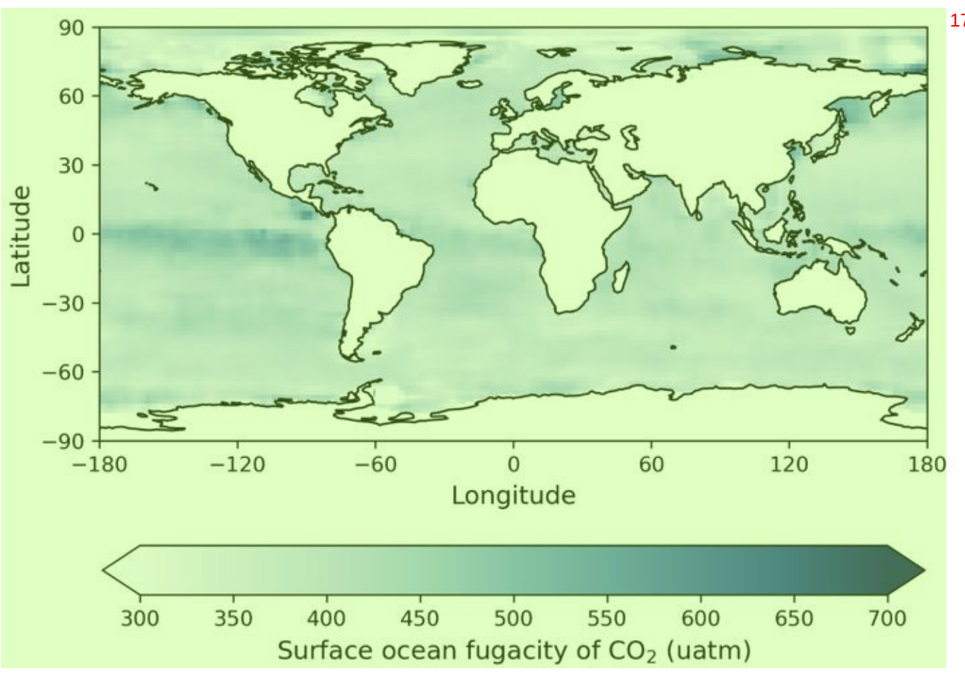
Model [12]


 [13]

Result [14]

Our models maintain accuracy with errors below a **5%** threshold(NRMSE), most models have losses below **2%** [15]


 [16]


 [17]

ICLR International Conference On Learning Representations

Our Paper

Our Website

Microsoft

Tsinghua University